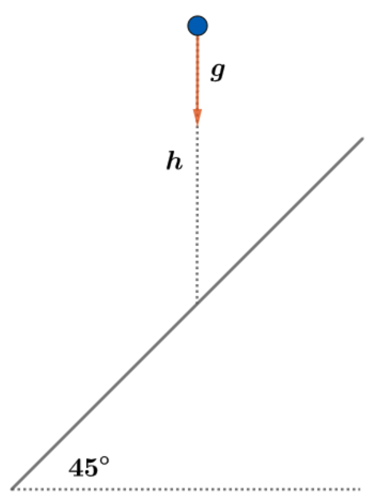
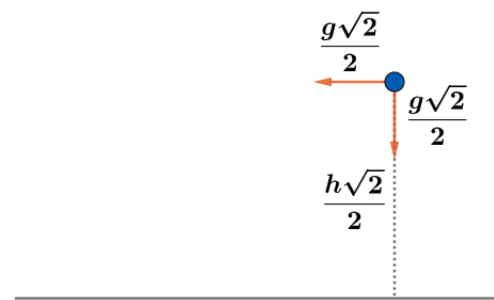
2022B F=ma Exam: Problem 1

Kevin S. Huang





We can rotate our coordinate system such that the inclined plane is horizontal. Now the ball is released at a height

$$h_0 = \frac{h\sqrt{2}}{2}$$

above the ground with horizontal and vertical accelerations

$$g_x = g_y = \frac{g\sqrt{2}}{2}$$

From kinematics, the time it takes to move between the highest point and the ground is

$$h_0 = \frac{1}{2}g_y T^2$$

$$T = \sqrt{\frac{2h_0}{g_y}} = \sqrt{\frac{2h}{g}}$$

The first impact point (at time $t = T$) is $d_1 = \frac{1}{2}g_x T^2$ from the starting horizontal position. The second impact point (at time $t = 3T$) is $d_2 = \frac{1}{2}g_x(3T)^2$ from the start:

$$\Delta d = d_2 - d_1 = \frac{1}{2}g_x(8T^2) = \frac{1}{2}\left(\frac{g\sqrt{2}}{2}\right)\frac{16h}{g} = 4\sqrt{2}h$$

so the answer is $\boxed{\text{D}}$.